

Assignment No. 1

Simple Text Generator Using Markov Chains

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Introduction

A Markov Chain-based text generator is a simple generative AI model that creates new text by learning patterns from existing text. Instead of understanding language like modern deep learning models, it uses probabilities of word sequences to produce sentences that resemble the training data.

Theory

A Markov Chain is a mathematical model that predicts the next state based only on the current state. This is known as the Markov Property, which assumes that future outcomes depend only on the present and not on past history. In text generation, each state represents a word (or a group of words), and transitions represent the probability of one word following another.

For example, if the word “machine” is often followed by “learning” in the training text, the model assigns a higher probability to that transition. These probabilities are calculated by analyzing how frequently word pairs or triplets occur.

Model Types

Unigram models choose words based only on frequency, with no context. Bigram models predict the next word using the current word. Trigram models use two previous words, producing more coherent sentences but requiring more data.

Working Principle

The system first processes training text and builds a probability table of word transitions. During generation, it starts with a word and repeatedly selects the next word based on learned probabilities. This process continues until a complete sentence is formed.

Advantages and Limitations

Markov models are simple, fast, and useful for learning basic natural language processing concepts. However, they do not understand meaning, often produce grammatically incorrect sentences, and struggle with long-range context.

Conclusion

A Markov Chain text generator demonstrates how probabilistic methods can be used in generative AI. While simple, it provides foundational insight into how machines can learn patterns in language and generate human-like text.